**Name:** ______________________ **Date:** ________________

# Introduction to Pythagoras' Theorem

Year 8 — Stage 4 Mathematics

**Learning Intentions**

- I can identify the hypotenuse in a right-angled triangle.
- I can use Pythagoras' theorem to calculate the hypotenuse.
- I can rearrange Pythagoras' theorem to find a shorter side.
- I can apply Pythagoras' theorem to a real-world situation.

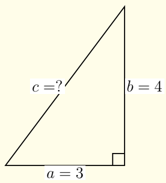
**Key Rule**

In a right-angled triangle, the **hypotenuse** $c$ (longest side, opposite the right angle) satisfies:

$$c^2 = a^2 + b^2 \qquad \Longrightarrow \qquad c = \sqrt{a^2 + b^2}$$

To find a **shorter side**: $a^2 = c^2 - b^2 \quad \Longrightarrow \quad a = \sqrt{c^2 - b^2}$

**Worked Example — Finding the Hypotenuse**




**Step 1:** Write the theorem.

$$c^2 = a^2 + b^2$$

**Step 2:** Substitute and evaluate.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

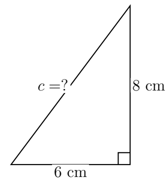
**Step 3:** Square root both sides.

$$c = \sqrt{25} = \mathbf{5}$$





**1.** [**Fluency**] Find the length of the hypotenuse $c$ in the triangle below. Show all working.




**2.** [**Misconception Check**] A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

**(a)** Find the length of the unknown shorter side $a$. Show full working.

**(b)** A student attempted the same problem and wrote:

$$a^2 = 5^2 + 13^2 = 25 + 169 = 194, \qquad \text{so} \quad a = \sqrt{194} \approx 13.9 \text{ cm.}$$

**Identify** the exact error the student made. Then write the correct solution.





**3.** [**Challenge — Real-World Application**] A 5 m ladder leans against a vertical wall. The foot of the ladder sits on level ground, 1.8 m from the base of the wall.

**(a)** Draw and label a right-angled triangle to represent this situation. Mark the hypotenuse, both known measurements, and the unknown height $h$.

**(b)** Calculate $h$, the height the ladder reaches up the wall. Give your answer correct to one decimal place.

**(c)** The foot of the ladder is then moved to 2.5 m from the wall. The ladder length stays the same.

Without fully recalculating, **explain** whether the ladder now reaches *higher* or *lower* up the wall, and why.





## Teacher Answer Key

*For teacher use only — not for student distribution*

**1. Fluency — Finding the hypotenuse (6–8–10 triple)**

$$\begin{aligned} c^2 &= 6^2 + 8^2 \\ &= 36 + 64 \\ &= 100 \\ c &= \sqrt{100} = \mathbf{10} \text{ cm} \end{aligned}$$

*Marking: 1 mark — correct substitution into* $c^2 = a^2 + b^2$. *1 mark — correct answer* $c = 10$ cm.

**2. Misconception Check — Finding a shorter side (5–12–13 triple)**

**(a)**

$$\begin{aligned} a^2 &= c^2 - b^2 \\ &= 13^2 - 5^2 \\ &= 169 - 25 \\ &= 144 \\ a &= \sqrt{144} = \mathbf{12} \text{ cm} \end{aligned}$$

*Marking: 1 mark — correct rearrangement* $a^2 = c^2 - b^2$. *1 mark — correct answer* $a = 12$ cm.

**(b)** The student **added** the squares ($5^2 + 13^2$) instead of **subtracting** ($13^2 - 5^2$). When finding a shorter side, the theorem must be rearranged to $a^2 = c^2 - b^2$. A further red flag: the student's answer ($\approx$13.9 cm) is *larger* than the hypotenuse (13 cm), which is geometrically impossible. Correct answer: $a = 12$ cm.

*Marking: 1 mark — correctly identifies addition error. 1 mark — corrected working and answer.*

**IMPORTED FROM CHAT:**
*[TEACHER ONLY] Refinement suggestion — add student-facing mark allocations. Currently no marks appear beside each part. For NSW Stage 4 exam readiness, consider annotating each part with (__/2) or similar so students practise gauging how much to write. Q2(b) in particular could become a 2-mark response question with an explicit instruction to give two distinct points (name the error; explain why the answer is impossible). This also doubles as a useful self-assessment prompt.*

**Name:** ____________________ **Date:** ______________

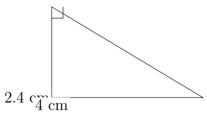
# Introduction to Pythagoras' Theorem

**Key Idea:** In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

$$c^2 = a^2 + b^2$$

**1.** A right-angled triangle has legs $a = 3$ cm and $b = 4$ cm. Find the length of the hypotenuse $c$.

**2.** The triangle below is right-angled at $B$. Use the diagram to find the length of the hypotenuse.




**3.** A student claims that if $a = 5$ and $b = 12$, then the hypotenuse is $c = 17$ because "you just add the sides". Explain why this is incorrect and calculate the correct value of $c$.





# Teacher Answer Key

**4. Solution:**

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = 5 \text{ cm}$$

**Marking guidance:** correct substitution; correct square/sum; correct square root.

**5. Solution:**

$$c^2 = 2.4^2 + 4^2 = 5.76 + 16 = 21.76$$

$$c = \sqrt{21.76} \approx 4.66 \text{ cm}$$

**Marking guidance:** diagram interpreted correctly; right angle identified; rounding acceptable to two decimal places.

**6. Solution:** The claim $c = a + b$ is incorrect because Pythagoras' Theorem uses squares, not addition.

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$c = 13$$

**Marking guidance:** clear explanation of misconception; correct working; correct final value.

**IMPORTED FROM CHAT:**
*— [TEACHER ONLY] — Consider adding one more representation-shift item (e.g., a rotated triangle) to strengthen transfer.*

**Name:** ____________________ **Date:** ______________

# Year 8: Introduction to Pythagoras' Theorem

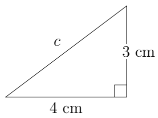
Pythagoras' Theorem applies to right-angled triangles.

$$a^2 + b^2 = c^2$$

where $c$ is the hypotenuse (the longest side).

**1. Getting Started**

The triangle below is right-angled.




Use Pythagoras' Theorem to find the value of $c$.

**2. Building Confidence**

A right-angled triangle has side lengths 5 cm and 12 cm.

Find the hypotenuse.





**3. Challenge**

The hypotenuse of a right-angled triangle is 13 cm.

One shorter side is 5 cm.

Find the length of the remaining side.





## Teacher Answer Key

**4.**

$$c^2 = 4^2 + 3^2$$

$$c^2 = 16 + 9 = 25$$

$$c = 5 \text{ cm}$$

**Marking guidance:** Substitutes correctly into the formula and calculates $c = 5$ cm.

**5.**

$$c^2 = 5^2 + 12^2$$

$$c^2 = 25 + 144 = 169$$

$$c = 13 \text{ cm}$$

**Marking guidance:** Identifies the hypotenuse and applies the theorem correctly.

**6.**

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12 \text{ cm}$$

**Marking guidance:** Rearranges correctly and solves for the unknown side.

**IMPORTED FROM CHAT:**

*— [TEACHER ONLY] — Add a fourth "spot the error" question where a student incorrectly calculates $3^2 + 4^2 = 7^2$ to explicitly target a common Year 8 misconception about squaring numbers.*

**Name:** ______________________ **Date:** ______________

# Year 8 Introduction to Pythagoras' Theorem

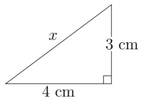
**Key Rule**
For a right-angled triangle with shorter sides $a$ and $b$, and longest side $c$:

$$a^2 + b^2 = c^2$$

The longest side opposite the right angle is called the hypotenuse.

**1. Starter**

Find the length of the hypotenuse.




Find $x$.

**2. Practice**

A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

Find the length of the other side.

**3. Challenge**

A ladder is leaning against a wall. The foot of the ladder is 6 m from the wall and the ladder itself is 10 m long.

How high up the wall does the ladder reach?

Give your answer in metres.





# Teacher Answer Key

**1.**

$$x^2 = 4^2 + 3^2$$

$$x^2 = 16 + 9$$

$$x^2 = 25$$

$$x = 5$$

**Answer:** 5 cm

Marking guidance:

- Substitutes correctly into Pythagoras' theorem.
- Squares numbers correctly.
- Gives answer with units.

**2.**

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12$$

**Answer:** 12 cm

Marking guidance:

- Identifies the hypotenuse.
- Rearranges correctly.
- Gives answer with units.





**3.**

$$10^2 = 6^2 + h^2$$

$$100 = 36 + h^2$$

$$h^2 = 64$$

$$h = 8$$

**Answer:** 8 m

Marking guidance:

- Forms the correct equation.
- Solves accurately.
- Includes units.

**IMPORTED FROM CHAT:**

*To draft the worksheet, I need one detail: what is the topic (and year/stage) you want the worksheet to cover? For example: Year 9 Maths – Linear Relationships, Stage 5 Science – Chemical Reactions, or Year 11 Maths Advanced – Trigonometric Functions.*

*— [TEACHER ONLY] — Consider adding a misconception-check question where students must identify the hypotenuse before applying Pythagoras' theorem, as Year 8 students commonly square the wrong side.*

**Name:** ____________________ **Date:** ______________

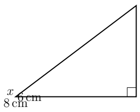
# Introduction to Pythagoras' Theorem

**Key Concept: Pythagoras' Theorem**

In any right-angled triangle, the square of the hypotenuse (the longest side, opposite the right angle) is equal to the sum of the squares of the other two sides.

$$c^2 = a^2 + b^2$$

**1. Access Starter (Finding the Hypotenuse)**
Calculate the length of the unknown side $x$ in the right-angled triangle shown below. Express your answer exactly, and then rounded to one decimal place.




**2. Fluency & Representation Shift (Finding a Shorter Side)**
A right-angled triangle has a hypotenuse of length 13 cm and one shorter side of length 5 cm.

(a) Sketch and label this triangle in the space below.

(b) Set up an equation and solve for the length of the remaining shorter side, $y$.





**3. Challenge: Application & Synthesis (Two-Step Problem)**
A ladder of length 5 m rests against a vertical wall. The base of the ladder is initially 3 m away from the base of the wall. If the top of the ladder slips down the wall by exactly 1 m, determine how far the base of the ladder slides outwards from its original position. Give your final answer to two decimal places.





# Teacher Answer Key & Marking Guide

**Sample Solutions & Marking Criteria**

**4. Solution:**

$$c^2 = a^2 + b^2$$
$$x^2 = 8^2 + 6^2$$
$$x^2 = 64 + 36 = 100$$
$$x = \sqrt{100} = 10\,\text{cm}$$

**Marking Guide:**

- **1 Mark:** Correct substitution into Pythagoras' Theorem.
- **1 Mark:** Correct final answer with appropriate units (cm).

**5. Solution:**

(a) Student sketch should show a right-angled triangle with the hypotenuse clearly labelled as 13 cm and one side as 5 cm.

(b) Calculation:

$$13^2 = y^2 + 5^2$$
$$169 = y^2 + 25$$
$$y^2 = 169 - 25 = 144$$
$$y = \sqrt{144} = 12\,\text{cm}$$

**Marking Guide:**

- **1 Mark:** Correctly drawn and labelled diagram.
- **1 Mark:** Correct algebraic manipulation to isolate $y^2$.
- **1 Mark:** Correct final value of 12 cm.

**6. Solution: Step 1: Find the initial height of the ladder ($h_1$).**

$$5^2 = 3^2 + h_1^2 \implies 25 = 9 + h_1^2 \implies h_1^2 = 16 \implies h_1 = 4\,\text{m}$$

**Step 2: Find the new height and new base distance.** The top slips down by 1 m, so the new height $h_2 = 4 - 1 = 3$ m. Let the new total distance from the wall to the base of the ladder be $d$.

$$5^2 = 3^2 + d^2 \implies 25 = 9 + d^2 \implies d^2 = 16 \implies d = 4\,\text{m}$$

**Step 3: Calculate how far it slid outwards.**

$$\text{Slid distance} = d - 3\,\text{m} = 4 - 3 = 1\,\text{m}$$

**Marking Guide:**





- **1 Mark:** Calculating the initial vertical height (4 m).
- **1 Mark:** Setting up the second scenario with the new height (3 m).
- **1 Mark:** Calculating the new base horizontal distance (4 m).
- **1 Mark:** Final subtraction yielding the correct sliding distance (1 m).

**IMPORTED FROM CHAT:**

*— [TEACHER ONLY] — Dynamic Refinement Suggestion: If you notice your Year 8s struggling to identify the hypotenuse visually before they even start calculating, ask me to add an introductory "Access" question where they simply highlight or trace the hypotenuse in three different, randomly rotated triangles without doing any math.*

**Name:** ______________________ **Date:** ________________

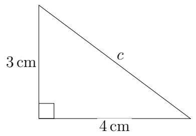
# Introduction to Pythagoras' Theorem

**Key idea:** In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$a^2 + b^2 = c^2$$

**1. Find the hypotenuse**

A right-angled triangle has side lengths 3 cm and 4 cm.




Find the length of the hypotenuse.

**2. Find a missing side**

A right-angled triangle has hypotenuse 13 cm and one side 5 cm.

Find the length of the other side.





**3. Challenge: Word problem**

A ladder leans against a wall. The base of the ladder is 6 m from the wall, and the ladder is 10 m long.

- How high up the wall does the ladder reach?





## Teacher Answer Key

**1.**

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$
$$c = \sqrt{25} = 5\,\text{cm}$$

**Answer:** 5 cm

**2.**

$$c^2 = a^2 + b^2$$
$$13^2 = 5^2 + b^2$$
$$169 = 25 + b^2$$
$$b^2 = 144$$
$$b = \sqrt{144} = 12\,\text{cm}$$

**Answer:** 12 cm

**3.**

$$c^2 = a^2 + b^2$$
$$10^2 = 6^2 + h^2$$
$$100 = 36 + h^2$$
$$h^2 = 64$$
$$h = \sqrt{64} = 8\,\text{m}$$

**Answer:** 8 m

**IMPORTED FROM CHAT:**

*I'm ready to build this, but I need a bit more detail first. What are the key inputs: subject (Maths/Science), year or stage, topic, number of questions, and desired difficulty or mode (e.g. exam-style, scaffolded, investigation)?*

*— [TEACHER ONLY] —*

*Add a misconception check (e.g. a triangle that is not right-angled) to ensure students do not apply Pythagoras indiscriminately.*

**Name:** ____________________ **Date:** ______________

# Introduction to Pythagoras' Theorem

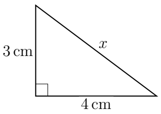
**Learning Intention**

To understand and apply Pythagoras' Theorem ($c^2 = a^2 + b^2$) to find missing side lengths in right-angled triangles. Remember that $c$ must always be the **hypotenuse** (the longest side, directly opposite the right angle).

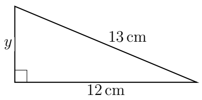
### 1. Fluency: Finding missing sides

For each of the right-angled triangles below, use Pythagoras' Theorem to calculate the value of the unknown side. Show all steps of your working.

**a)** Find the hypotenuse, $x$.




**b)** Find the shorter side, $y$.




### 2. Application: The Farm Gate

A farmer is building a rectangular wooden gate that is 2.4 m wide and 1.0 m high. To ensure the gate does not sag, she needs to attach a diagonal brace across the back.

(a) Draw a simple, labelled diagram to represent this scenario.

(b) Calculate the exact length of the diagonal piece of wood required.

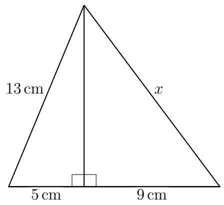




### 3. Challenge: The Composite Shape

The diagram below shows two right-angled triangles joined together. Calculate the length of the side marked $x$.

*Hint: You will need to find the length of the shared middle side first.*




**IMPORTED FROM CHAT:**

*I am ready to design your LaTeX worksheet. Could you please provide your teacher brief, including the subject (Maths, Science, or Integrated STEM), Year/Stage, topic, and preferred pedagogy mode?*

*— [TEACHER ONLY] — Refinement Suggestion: If you notice students struggling with identifying whether to add or subtract squares, I can add a quick "Misconception Check" diagnostic question before the fluency task where they must spot and explain the error in a provided, flawed calculation.*





# Teacher Answer Key

### Question 1: Fluency

- **1a)** Using $c^2 = a^2 + b^2$
  $x^2 = 3^2 + 4^2$
  $x^2 = 9 + 16$
  $x^2 = 25$
  $x = \sqrt{25} = 5\,\text{cm}$
  *Marking guide: 1 mark for correct substitution, 1 mark for correct final answer with units.*
- **1b)** Using $a^2 + b^2 = c^2$
  $y^2 + 12^2 = 13^2$
  $y^2 + 144 = 169$
  $y^2 = 169 - 144$
  $y^2 = 25$
  $y = \sqrt{25} = 5\,\text{cm}$
  *Marking guide: 1 mark for identifying the hypotenuse is 13 and setting up the subtraction/equation correctly, 1 mark for final answer.*

### Question 2: Application

- **2a)** Students should draw a rectangle with base 2.4 m, height 1.0 m, and a diagonal drawn in. A right-angle symbol should be present in the corner.
- **2b)** Using $c^2 = a^2 + b^2$
  $c^2 = 2.4^2 + 1.0^2$
  $c^2 = 5.76 + 1.0$
  $c^2 = 6.76$
  $c = \sqrt{6.76} = 2.6\,\text{m}$
  *Marking guide: 1 mark for diagram, 1 mark for calculation, 1 mark for final answer in metres.*

### Question 3: Challenge

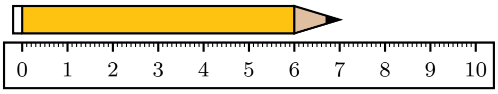
- **Step 1: Find the shared vertical side (let's call it $h$)**
  Looking at the left triangle:
  $h^2 + 5^2 = 13^2$
  $h^2 + 25 = 169$
  $h^2 = 144$
  $h = \sqrt{144} = 12\,\text{cm}$
- **Step 2: Find $x$**
  Looking at the right triangle:
  $x^2 = h^2 + 9^2$
  $x^2 = 12^2 + 9^2$
  $x^2 = 144 + 81$
  $x^2 = 225$
  $x = \sqrt{225} = 15\,\text{cm}$
  *Marking guide: 1 mark for finding the intermediate side (12cm), 1 mark for setting up the second equation, 1 mark for correct final answer.*

**Name:** ____________________ **Date:** ________________

# Cross-Subject Worksheet Capability Sampler

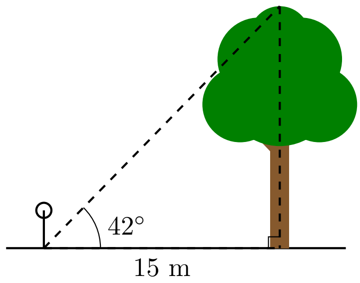
This sampler demonstrates the engine's ability to generate printable worksheet structures across subjects, diagrams, answer spaces, and teacher answer keys. A normal teacher request would generate one coherent worksheet for a specific class, topic, and learning goal.

**1.** A student is using a ruler to measure the length of a pencil. Observe the diagram below and state the exact length of the pencil in centimetres. **[1 Mark]**

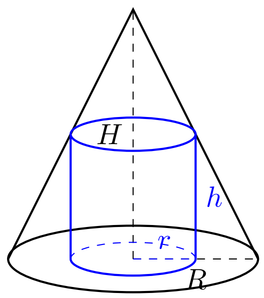



______________________________________________

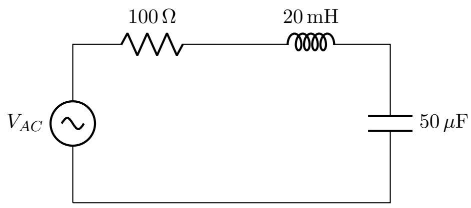
**2.** A person standing 15 m away from the base of a tree uses a clinometer to measure the angle of elevation to the top of the tree, finding it to be 42°. Assuming the measurement is taken from ground level for simplicity, calculate the height of the tree to two decimal places. **[2 Marks]**




**3.** A cylinder of radius $r$ and height $h$ is inscribed inside a right circular cone of base radius $R$ and height $H$. Find the maximum possible volume of the inscribed cylinder in terms of $R$ and $H$. **[4 Marks]**




**4. Physics (Electromagnetism):** Consider the series RLC circuit below, driven by an AC voltage source. Calculate the resonant frequency of this circuit in Hertz. **[3 Marks]**

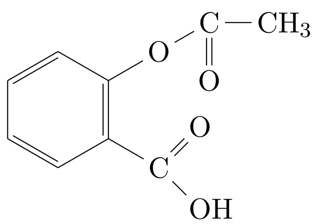



______________________________________________

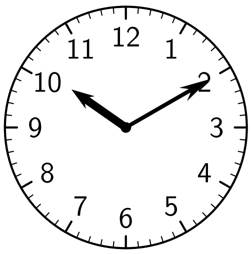
______________________________________________

______________________________________________

**5. Advanced Chemistry (Organic Synthesis):** Below is the skeletal structure of Aspirin. Identify the three primary functional groups present. Write the balanced chemical equation for the synthesis of Aspirin reacting salicylic acid with acetic anhydride. **[4 Marks]**

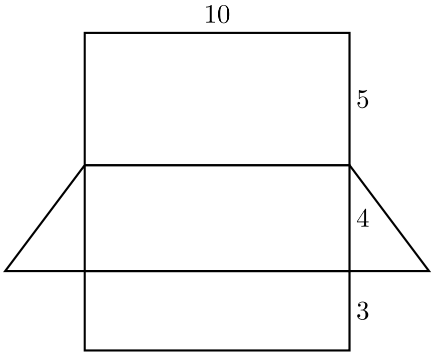


**6. Junior Scale:** The time shown below is in the morning. Write the current time in 12-hour and 24-hour formats. **[2 Marks]**



**12hr:** ________ **24hr:** ________

**7.** The diagram below shows the 2D net of a right triangular prism. Given the dimensions in cm, calculate the total surface area. **[3 Marks]**

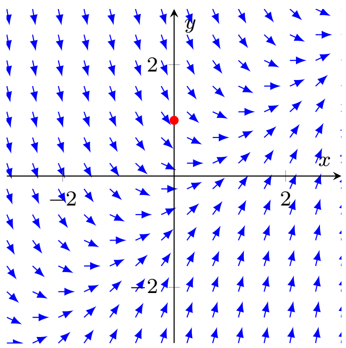



______________________________

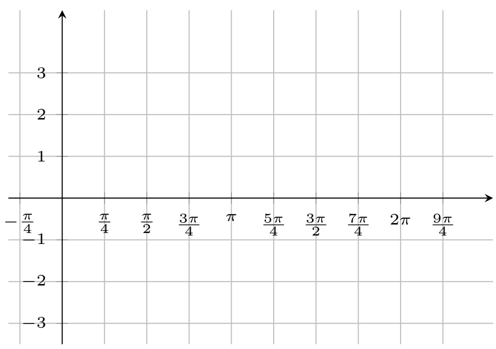
______________________________

______________________________

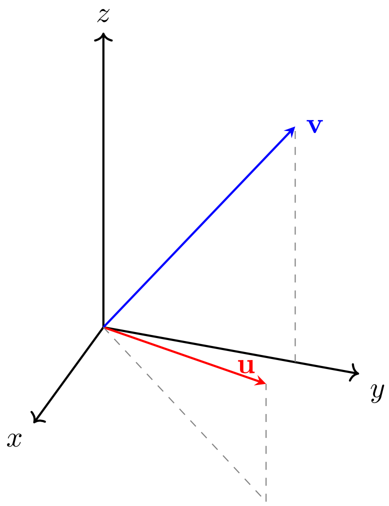
**8.** Consider the differential equation $\frac{dy}{dx} = x - y$. A direction field is provided below. Sketch the solution curve through $(0, 1)$. **[3 Marks]**




**9.** A wave function is modelled by $y = 2\sin(x) + 1$. Sketch the graph of this function bounding the $x$-axis from $-\frac{\pi}{4}$ to $\frac{9\pi}{4}$. **[3 Marks]**




**10. Mathematics Ext 1 (3D Vectors):** Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = 4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 0\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$. Find the exact angle between them to the nearest degree. **[3 Marks]**

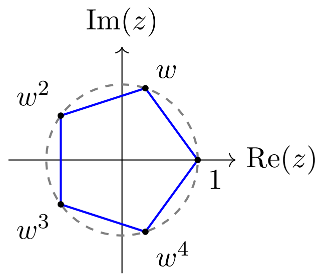



______________________________

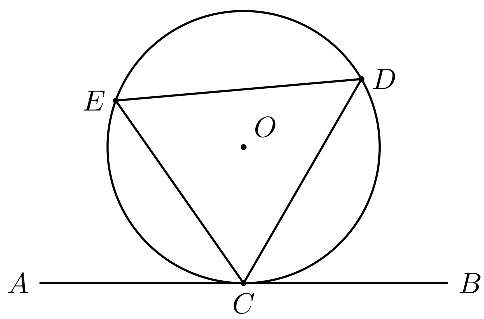
______________________________

______________________________

**11. Mathematics Ext 2 (Complex Roots):** The diagram illustrates the 5th roots of unity. Let $w = \text{cis}\left(\frac{2\pi}{5}\right)$. Prove that $1 + w + w^2 + w^3 + w^4 = 0$. **[3 Marks]**




**12. Geometry (Circle Theorems):** $AB$ is a tangent to the circle at $C$. If $\angle BCD = 60°$, calculate $\angle CED$. Give reasons. **[2 Marks]**

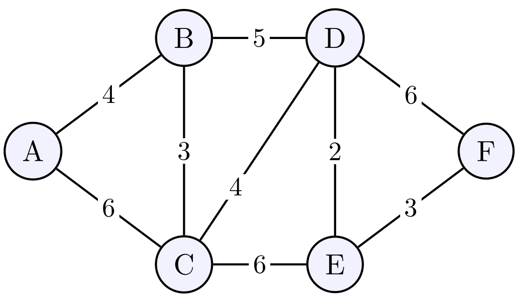


**13. Statistics:** The discrete random variable $X$ has the probability distribution below. Determine $k$, and calculate $E(X)$. **[3 Marks]**

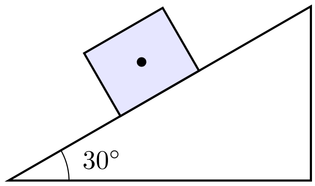
| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.1 | $2k$ | 0.3 | $k$ | 0.15 |

**14. Computer Science:** A sequence used in an encryption algorithm is defined recursively as $a_n = 2a_{n-1} - 1$, with seed $a_1 = 3$. Calculate $a_4$. **[2 Marks]**

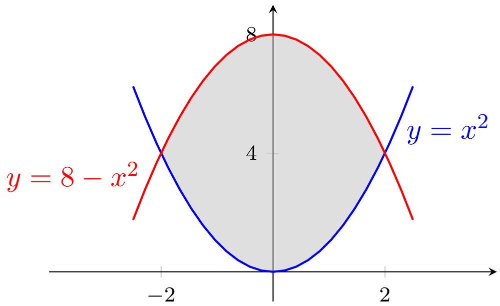
**15. Mathematics Standard 2 (Networks):** The network diagram below represents the travel times in minutes between six towns, labelled A to F. Determine the shortest time taken to travel from town A to town F, and list the vertices in this shortest path. **[2 Marks]**



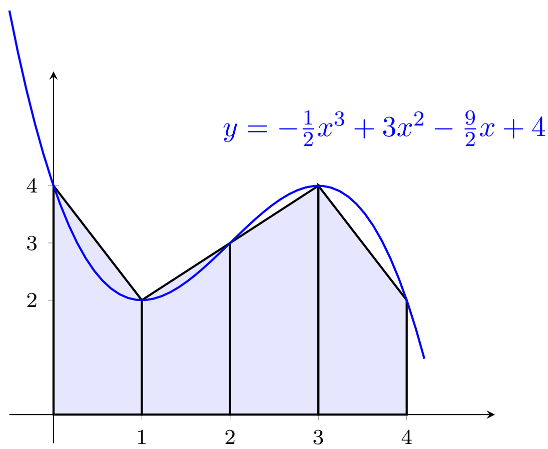

**16. Physics (Mechanics):** A block of mass $m = 5$ kg rests on an inclined plane at an angle of $\theta = 30°$. Draw a free-body diagram on the block showing the normal force, gravity, and friction. **[3 Marks]**




**17. Calculus (Integration):** Find the exact area of the shaded region bounded by the curves $y = x^2$ and $y = 8 - x^2$. **[3 Marks]**




**18. Calculus (Numerical Integration):** The diagram below shows the curve $y = -\frac{1}{2}x^3 + 3x^2 - \frac{9}{2}x + 4$. Use the Trapezoidal Rule with 4 subintervals to find an approximation for the area bounded by the curve, the $x$-axis, and the lines $x = 0$ and $x = 4$. **[3 Marks]** $\int_a^b f(x)\,dx \approx \frac{h}{2}\left[f(a) + 2\sum_{i=1}^{n-1} f(x_i) + f(b)\right]$



**19. English Advanced (Literary Analysis):** Translate the following excerpt from Shakespeare's *Hamlet* into modern Australian vernacular. Additionally, identify and analyse the primary literary device used by Shakespeare in the final two lines to convey mental anguish. **[4 Marks]**

**William Shakespeare, *Hamlet***

To be, or not to be, that is the question:
Whether 'tis nobler in the mind to suffer
The slings and arrows of outrageous fortune,
Or to take arms against a sea of troubles...

**20. Modern History (Timeline Analysis):** The timeline below maps four critical flashpoints of the Cold War. Select ONE of these events and evaluate its long-term impact on United States-Soviet Union geopolitical relations. **[4 Marks]**

| **1948** | **1962** | **1979** | **1989** |
|---|---|---|---|
| Berlin Blockade | Cuban Missile Crisis | Soviet-Afghan War | Fall of Berlin Wall |

**21. Spot the Error (Calculus):** **[1 Mark]**

A student was asked to find the volume of the solid generated when the region bounded by $y = x^2$, the x-axis, and $x = 3$ is rotated around the x-axis using the Disk Method. They submitted the following step-by-step working. Identify the specific line number where the logical error occurred, and explain the mistake.

**Student's Working**

**Line 1:** $V = \pi \int_0^3 r(x)\, dx$
**Line 2:** $V = \pi \int_0^3 (x^2)\, dx$
**Line 3:** $V = \pi \left[\frac{x^3}{3}\right]_0^3$
**Line 4:** $V = 9\pi$ units$^3$

 Teacher draft: verify, adapt, then distribute

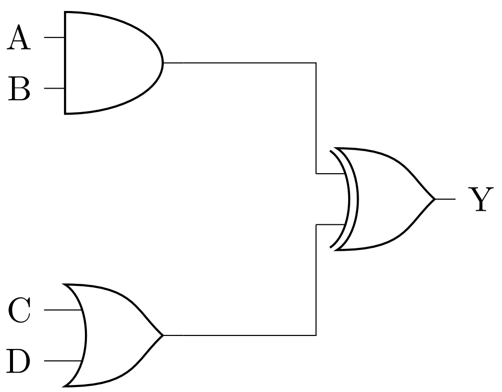
**22. Parameter Shift (Ecology):** **[3 Marks]**

Consider a standard sinusoidal predator-prey population model. If a highly aggressive invasive species is suddenly introduced to the ecosystem that competes exclusively with the prey (reducing the prey's carrying capacity significantly), explain *without doing any calculations* how the amplitude and phase shift of the predator's population wave will be fundamentally altered over time.

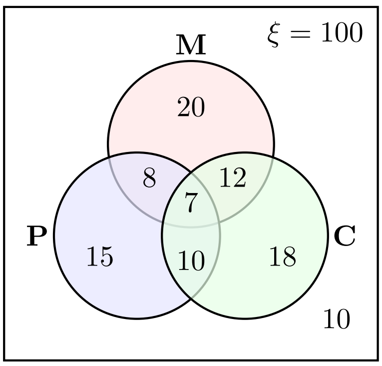
**23. Strategic Management (Business Studies):** **[3 Marks]**

A legacy automotive company has invested $500 million into developing a highly efficient internal combustion engine. Before completion, a breakthrough in solid-state batteries makes electric vehicles significantly cheaper to produce. The board argues they must launch the new combustion engine to "recover our investment." Identify the logical fallacy driving the board's argument and explain why proceeding could harm their long-term market position.

**24. Computer Science (Digital Logic):** Write the Boolean expression for the output $Y$ of the following logic circuit. Construct the corresponding truth table for the four inputs A, B, C, and D. **[3 Marks]**




**25. Statistics (Set Theory):** A survey of 100 students asked if they study Maths (M), Physics (P), or Chemistry (C). The results are mapped in the Venn diagram below. Calculate the exact number of students who study *only one* of these subjects, and the probability that a randomly selected student studies Physics given that they study Maths. **[3 Marks]**



# Teacher Answer Key

### Q1: Measurement
7.0 cm (Pencil tip aligns exactly with the 7 marker).

### Q2: Trigonometry
$\tan(42°) = \frac{h}{15}$
$h = 15 \times \tan(42°) \approx 13.51$ m.

### Q3: Optimization
By similar triangles: $\frac{h}{H} = \frac{R-r}{R} \implies h = H\left(1 - \frac{r}{R}\right)$.
$V = \pi r^2 H\left(1 - \frac{r}{R}\right)$.
$\frac{dV}{dr} = \pi H\left(2r - \frac{3r^2}{R}\right) = 0 \implies r = \frac{2}{3}R$.
$h = \frac{1}{3}H$. Max Vol $= \frac{4}{27}\pi R^2 H$.

### Q4: RLC Circuit
$f_r = \frac{1}{2\pi\sqrt{LC}}$. $L = 20 \times 10^{-3}$ H, $C = 50 \times 10^{-6}$ F.
$f_r = \frac{1}{2\pi \times 10^{-3}} = \frac{1000}{2\pi} \approx 159.15$ Hz.

### Q5: Chemistry
Groups: Carboxyl (carboxylic acid), Ester, Aromatic (benzene) ring.
Eq: $C_7H_6O_3 + C_4H_6O_3 \longrightarrow C_9H_8O_4 + C_2H_4O_2$

### Q6: Time
12-hour: 10:10 AM. 24-hour: 10:10.

### Q7: Surface Area Net
Rectangles: $(10 \times 3) + (10 \times 4) + (10 \times 5) = 120$ cm$^2$.
Triangles: $2 \times (\frac{1}{2} \times 3 \times 4) = 12$ cm$^2$.
Total SA $= 132$ cm$^2$.

### Q8: Direction Field
Solution curve passes through $(0, 1)$, moves downwards to the right (negative gradient), and eventually becomes asymptotic to the line $y = x - 1$.

### Q9: Trig Graphing
Standard sine wave properties: amplitude $= 2$, vertical shift $= +1$, range $[-1, 3]$. The curve starts at $(0, 1)$, peaks at $(\frac{\pi}{2}, 3)$, returns to the midline $y = 1$ at $x = \pi$, and troughs at $(\frac{3\pi}{2}, -1)$.

### Q10: 3D Vectors
$\mathbf{u} \cdot \mathbf{v} = (4)(0) + (4)(3) + (2)(4) = 20$.
$|\mathbf{u}| = 6$, $|\mathbf{v}| = 5$.
$\cos\theta = \frac{20}{30} \implies \theta \approx 48°$.

### Q11: Complex Roots
Roots are solutions to $z^5 - 1 = 0$.
Geometric sum $S_5 = \frac{1(1-w^5)}{1-w}$.
Since $w^5 = 1$, sum is 0.

### Q12: Circle Geometry
$\angle CED = 60°$. Reason: Alternate Segment Theorem.

### Q13: Statistics
Sum of probabilities $= 1 \implies 3k + 0.55 = 1 \implies k = 0.15$.
$E(X) = (0)(0.1) + (1)(0.3) + (2)(0.3) + (3)(0.15) + (4)(0.15) = 1.95$.

### Q14: Algorithmic Recursion
$a_1 = 3 \implies a_2 = 5 \implies a_3 = 9 \implies a_4 = 17$.

### Q15: Networks
Shortest path: $A \to B \to D \to E \to F$. Total time $= 4 + 5 + 2 + 3 = 14$ minutes.

### Q16: Inclined Plane
A correct free-body diagram should show weight $mg$ vertically downward, normal force perpendicular to the plane, and friction parallel to the plane opposing likely motion down the slope.

### Q17: Area Between Curves
Intersections: $x^2 = 8 - x^2 \Rightarrow x = \pm 2$.
Area $= \int_{-2}^{2} [(8 - x^2) - x^2]\, dx = \int_{-2}^{2} (8 - 2x^2)\, dx = \frac{64}{3}$ square units.

### Q18: Numerical Integration
$h = 1$. Function values: $f(0) = 4$, $f(1) = 2$, $f(2) = 3$, $f(3) = 4$, $f(4) = 2$.
Area $\approx \frac{1}{2}[4 + 2(2 + 3 + 4) + 2] = 12$ square units.

### Q19: English
Accept any accurate modern paraphrase preserving the existential conflict. Device: metaphor, especially "slings and arrows" and "sea of troubles", used to represent suffering and overwhelming mental anguish.

### Q20: History
Open response. Strong answers evaluate one event's long-term geopolitical impact, e.g. the Cuban Missile Crisis increased nuclear caution, produced direct communication channels, and shaped later arms-control thinking.

### Q21: Spot the Error
First error: Line 1. The Disk Method requires $V = \pi \int [r(x)]^2\, dx$, not $\pi \int r(x)\, dx$. Line 2 repeats the same error.

### Q22: Ecology Parameter Shift
A lower prey carrying capacity should reduce prey peaks, which then reduces predator peaks after a delay. Predator oscillations may show

lower amplitude and a shifted lag because predator numbers respond after prey availability changes.

**Q23: Business Studies**
Logical fallacy: sunk cost fallacy. Past spending cannot justify future investment if market conditions have changed. Launching the engine may divert resources from EV competitiveness and damage long-term positioning.

**Q24: Digital Logic**
$Y = (A \wedge B) \oplus (C \vee D)$. Full truth table should evaluate $A \wedge B$ and $C \vee D$ first, then XOR the two intermediate outputs.

**Q25: Set Theory**
Only one subject: $20 + 15 + 18 = 53$.
$P(P \mid M) = \frac{n(P \cap M)}{n(M)} = \frac{8+7}{20+8+12+7} = \frac{15}{47}$.

**ISNSW AI Agent Competition**